



Genomic medicine and personalized treatment: a narrative review

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Abstract

Genomic medicine, which integrates genomics and bioinformatics into clinical care and diagnostics, is transforming healthcare by enabling personalized treatment approaches. Advances in technologies such as DNA sequencing, proteomics, and computational power have laid the foundation for individualized therapies that account for genetic variations influencing disease risk, progression, and treatment response. This review explores the historical milestones leading to current applications of genomic medicine, such as targeted therapies, gene therapies, and precision medicine, in fields including cardiovascular diseases, oncology, and rare genetic disorders. It highlights the use of next-generation sequencing and third-generation sequencing to improve diagnostic accuracy and treatment outcomes, emphasizing the role of genomic data in advancing personalized treatments. Furthermore, emerging therapies such as CRISPR/Cas-based genome editing and adeno-associated viral vectors showcase the potential of gene therapy in addressing complex diseases, including rare genetic disorders. Despite promising advancements, challenges remain in fully integrating genomic medicine into routine clinical practice, including cost barriers, data interpretation complexities, and the need for widespread genomic literacy among healthcare professionals. The future of genomic medicine holds transformative potential for revolutionizing the diagnosis, treatment, and management of both common and rare diseases.

Keywords: cardiology, genomic medicine, genomics

Introduction

Genomic medicine refers to genomics and bioinformatics in the context of clinical care and diagnostics^[1]. The Human Genome Project was an international collaboration with respect to research that attempted to study the entirety of what is known as the human genome. The human genome is roughly 6 billion

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HIGHLIGHTS

- Genomic medicine integrates genomics and bioinformatics into clinical care and diagnostics, ushering in the era of personalized medicine.
- With roots in the groundbreaking discoveries of Watson and Crick in 1953 and the Human Genome Project, advancements in DNA sequencing have revolutionized our understanding of human genetic variation and its role in health and disease.
- From early RNA sequencing methods to cutting-edge third-generation sequencing, these innovations have enabled longer and more accurate DNA reads, paving the way for tailored treatments. Personalized medicine leverages genomic and biochemical data to address interindividual variations, optimizing therapeutic outcomes.

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- Despite its promise, challenges remain in improving patient characterization and creating effective, individualized treatments, highlighting the ongoing need for innovation in genomic medicine. This review evaluates the transformative impact, current applications, and challenges of genomic medicine in advancing personalized care.

DNA base pairs in size and to put it succinctly, contains all the code needed to create what we can call a human being. The fact that different DNA variants being dispersed throughout this genome is what makes an individual different from the rest. Likewise, these DNA variants might also be responsible for causing pathologies that manifest during one's lifetime. Some of these variants can be directly responsible for these pathologies while other variants might be an indicator of how an individual's body will react to a certain treatment. This is where personalized medicine tries to enhance the current scope of medicine^[2]. In 1953, Watson and Crick published their first paper on the double helix structure of DNA^[3]. In the same year, the sequencing of a biological molecule was completed for the first time via a refined partition chromatography method^[4]. At the end of the 1960s, RNA sequencing was still ahead of DNA sequencing^[5]. By 1979, the idea of "shotgun sequencing" was proposed which uses bacterial vectors to clone fragments of a DNA molecule, a procedure allowing sequencing of longer DNA molecules in less time^[6]. In 1984, the genome of the Epstein-Barr B95-8 strain was determined^[7]. Here-on, a myriad of full-genome screening project was launched and succeeded^[5]. GenBank, the US National Institute of Health (NIH) sequence database, was founded in 1982^[5]. Advancements in microfabrication, imaging, and computational power led to new sequencing methods. These involve preparing a DNA library by fragmenting DNA, attaching adapters, amplifying it, and then sequencing on a flow cell using massive parallel sequencing^[8]. Beginning in the 2010s, third-generation sequencing emerged with the ability to sequence single DNA molecules without amplification. These technologies now produce much longer reads than next-generation sequencing (NGS), ranging from several to hundreds of kilobase pairs^[5]. Personalized medicine tailors treatment based on individual patient data, such as genomic and biochemical information, due to significant inter-individual variations. Advances in technologies like DNA sequencing and proteomics have highlighted the need for this approach. Future challenges include enhancing the efficiency of patient characterization and developing effective personalized treatments, although universally effective drugs may still be sought but harder to find^[9]. The purpose and objective of this review is to explore the role of genomic medicine in advancing personalized treatment and to assess its current applications, benefits, and challenges.

Discussion

Genetics and genomic principles

DNA, genes, and genomes constitute the fundamental structural components of an organism's biological framework. DNA double

helix with structural base pairing is the most widely recognized DNA structure. It is evident from this structure that DNA is structurally dynamic and capable of adopting alternative secondary structures^[3]. A genome is an organism's complete set of DNA sequences. Although people in this world may look different, all human genomes are highly similar^[10]. It includes all of an organism's genes and non-coding sequences. Most genomes consist of a linear polymer of DNA wrapped around octameric histone protein complexes to generate a chromatin structure resembling beads on a string^[11].

Genetic variations are the changes in the DNA sequences that range from single nucleotide changes to large structural alterations. Some nonhuman genetic variations are closely related to certain diseases or individual patient responses to certain medications^[12], signifying the need of specific treatment options. These variations could be single nucleotide polymorphisms (SNPs); the simplest form of DNA variation which may influence promoter activity (gene expression), messenger RNA (mRNA) conformation (stability), and subcellular localization of mRNAs and/or proteins and hence may produce disease^[13], short insertions and deletions (INDELs); the second most common type of genetic variations^[14] characterized by addition and removal of small nucleotide sequences within the genome, copy number variations (CNVs) that arise from genomic rearrangements, primarily owing to deletion, duplication, insertion, and unbalanced translocation events^[15] etc. Some human genetic variations are closely related to certain diseases or individual patient responses to certain medications^[12], which makes it possible to opt for the treatment that brings the best outcome for the patients. For example, in precision medicine, physicians can choose different medications to help their patients quit smoking by examining the patient's speed of nicotine metabolism^[16]. Recent studies on genetic variation have moved from examining genes tied to rare singlegene disorders such as cystic fibrosis to investigating genes involved in multifactorial diseases such as cancer and cardiovascular disorders. Therefore, studying genetic variations is not only enriching our knowledge of different disease mechanisms but is also modifying our diagnostic and therapeutic approaches.

Integration of genomic data into clinical practice

It is a slow process yet retreat in knowledge is increasing the use of genomic data and genomic medicine in clinical care^[17]. Advancement in genetics brings genetic medicine and genetic data into clinical practice improving the diagnosis of rare diseases, illness related risk improvement, and treatment efficiency through advanced measurement and methods^[18]. Nextgeneration sequencing (NGS) has changed the genomics and not only improve the method but also lowering the costs, can perform rapid genome sequencing and has several medical uses^[19].

Genetic testing is important for the detection of inherited and acquired disorders, and also for treatment responses. Multiple genetic tests are used including targeted single-gene assays, gene panels, whole-exome sequencing, and whole-genome sequencing. Chromosomal testing use for detecting changes in chromosomes like additional or missing copies and any large segment

modifications^[20]. Exome sequencing improves genetic diagnosis and aid in the prenatal identification of structural abnormalities or genetic disorder. Combining copy number variant and single nucleotide variant analyses increases accuracy, whereas lowpass genome sequencing provides higher resolution^[21]. Combining copy number variant sequencing and karyotyping improves the identification of prenatal pathogenic chromosomal abnormalities, enhancing the accuracy of prenatal diagnosis^[22].

Fluorescence in situ hybridization (FISH) used for detection of tumor-specific genetic variations, enhancing diagnosis and treatment^[23]. Genetic testing for prostate cancer, especially in metastatic patients, reveals up to 15% of germline mutations. Pre-test counseling covers inherited risk, diagnostic scope, results, and management options, enhancing personalized care with precision medicine^[24]. Myeloid neoplasms and acute leukemias resulting from somatic mutations are helped by enhanced genomic testing, like whole-genome sequencing, for accurate diagnosis and evaluation of risk, thus improving personalized treatment and clinical decision-making^[25].

Adequate care for epilepsy is difficult due to numerous syndromes and unique responses, however current genetic discoveries have found abnormalities in ion channels and neurotransmitter receptors in many individuals, whole-exome and whole-genome sequencing methods have enhanced our knowledge and led to precision treatment for particular diseases, like Dravet syndrome, pyroxidine-dependent epilepsy, and glucose transporter 1 deficiency^[26].

Gene therapy in cardiovascular diseases

Gene diagnosis in cardiovascular diseases is gaining attention, especially monogenic cardiovascular diseases. These are the diseases that have cardiovascular damage as their phenotype, e.g., cardiomyopathies, cardiac ion channel disease (long QT syndrome, Brugada syndrome, PVTs), inherited hypertension, inherited aortic diseases^[27].

There has been established a causal link between risk of DNA methylation at cpG site and various subtypes of CVD, prior MI, atherosclerotic disease in a recent epigenome wide association study (EWAS)^[28]. Selenium supplementation has been known to inhibit DNMT2 mediated DNA methylation of glutathione peroxidase 1 gene promoter in cardiomyocytes reducing the reactive oxygen species and toxicity to cardiomyocytes, and thus protecting the heart during its failure^[29].

Genetics play an important role in cardiomyopathies. Pathogenic variants in MYH7 gene, MYBPC3 gene are the most common in encoding abnormal sarcomeric proteins causing Hypertrophic cardiomyopathy. TTN gene, LMNA gene are the most commonly implicated genes in dilated cardiomyopathy. DSC2, DSG2 genes are implicated in arrhythmogenic right ventricular cardiomyopathy. Pathological variant genes testing is implicated to improve prognosis via early screening. Screening of first-degree relatives is also implicated via serial ECGs and echocardiography^[30].

Familial hypercholesterolemia is associated with genes such as LDLR, APOB, PCSK9, and APOE. Most of them have autosomal

dominant variants and increase the risk of coronary artery disease, atherosclerotic disease, peripheral arterial disease. Early genetic detection can modify the course of disease by early interventions like lifestyle modifications, exercise, blood pressure control, early initiation of statins, and PCSK9 inhibitors^[31]. Rather than genome sequencing or exome sequencing, capillary electrophoresis sequencing or next-generation sequencing targeted to known FH gene variants provide a more comprehensive result^[32].

Molecular pathophysiology of cardiac diseases can encourage preclinical gene therapy. Adeno-associated viral vector helps in introducing therapeutic genes in heart. Sarcoplasmic reticulum Ca²⁺ ATPase protein delivery has shown promising result in phase 1 trials to improve cardiac function in heart failure. Crispr/Cas based genome engineering has gained wide recognition for treating cardiovascular disease^[33].

Ultrasound targeted micro-bubble (UTM) strategy has gained recognition. Particularly, lipid micro-bubble carrying VEGF and stem cell factor has shown to improve myocardial perfusion and ventricular function in patients with MI^[34].

Hypertrophy of ventricles has been shown to reverse with UTM mediated delivery of miR-133 in cardiomyocytes^[35]. Anterograde arterial infusion has been indicated in patients with unstable and advanced heart failure, retrograde infusion in patients with impaired coronary artery circulation and limited potential for revascularization and direct intramyocardial infusion for focal arrhythmia therapy. Intra-coronary delivery of Ad vector encoding beta2 AR percutaneously, in rabbits, has shown to improve global ventricular systolic and contractility performance^[33].

Applications of genomic medicine in oncology

Over the past decade recent advances in genomic medicine has enhanced diagnosis and management of neoplastic diseases by knowing underlying molecular process. Cancer genomic profiling has shown to detect gene amplification, gene deletion mutation, gene fusion of the target genes. These results are interpreted extensively and reflected on treatments. Examples of genomic profiling tests are OncoGuide™ NCC Oncopanel System, FoundationOne® CDx Cancer Genomic Profile, TCGA OncoPanel, OncoPrint™ Target Test System. The system used for these tests also functions as a companion diagnostic^[36].

Targeted therapies and immunotherapies in oncology like monoclonal antibody against HER 2 which is overexpressed in HER2 positive breast cancer has revolutionized treatment^[37]. Osimertinib targeted against EGFR-mutated non-small cell lung cancer has drastically improved disease-free survival^[38]. In advanced melanoma, Ipilimumab, a monoclonal antibody directed against cytotoxic T-lymphocyte antigen (CTLA-4) has improved survival^[39].

Denosumab targeted against nuclear factor κ-B ligand RANK-L, inhibits osteoclastic activation and prevents further growth in giant cell bone tumor^[40]. Avelumab, a PD-1 inhibitor, is successfully used in Merkel cell carcinoma, Urothelial carcinoma and Renal cell carcinoma^[41].

Targeted gene therapy can help destroy tumor without being aggressive with therapies. A case report on 56 years old man with lung adenocarcinoma.

Patient's PD-L1 TPS was 70% and patient was started on pembrolizumab but recurrence was evidenced after 6th cycle^[42]. Other therapies also failed and led to side effects. Thereafter, the patient was enrolled in clinical study conducted by Japanese advanced medical treatment system and was found to be positive EGFR L858R-K860I doublet mutation. Treatment with oral osimertinib led to partial remission in just one month. Patient tolerated this drug and no side effects were noted^[43].

Sometimes targeted therapy can lead to new gene activation and neoplastic transformation. Targeted therapy against triple negative breast cancer can lead to the development of metastatic malignant melanoma^[44]. Secondary tumors and T-cell lymphoma can occur after CAR T-cell therapy. For patients who received axicabtagene ciloleucel therapy for diffuse large b cell lymphoma developed lethal T cell lymphoma^[45]. Furthermore, In recent studies rigid extracellular matrix of cells (ECM) has demonstrated increased tumorigenesis. Targeting ECM stiffness, can lead to collagen depletion and has emerged as potential cancer therapy^[46].

Rare genetic disorders

There are a wide variety of rare unexplained genetic disorders and developmental anomalies. Their apt diagnosis can help the patients understand their condition better. Recent advancements with NGS, which includes whole genome sequencing, whole exome sequencing, whole mtDNA sequencing, targeted exome sequencing and RNA sequencing, has countered the limitations of more traditional methods like Sanger method, karyotyping, and chromosomal arrays for rare genetic diseases^[46].

Whole exome sequencing was done on ~500 000 individuals in UK Biobank that identified about 564 distinct genes that had significant trait associations, e.g., CHD2 with chronic lymphocytic leukemias of B-cell type, COL1A1 with bone disorders, SERPINC1 with coagulation defects, etc.^[47,48].

Many Mitochondrial disorders have been linked with mutations in mitochondrial DNA or nuclear DNA using next-generation sequencing, e.g., MT-TL1, MT-TN mutation causing progressive external ophthalmoplegia, Kearns-Sayre syndrome (KSS) caused by single large-scale deletion^[49]. A whole exome sequencing (WES) analysis was conducted in Lebanon for neurological diseases in consanguineous families^[50]. Thirty-three gene variations were identified among the pre-screened consanguineous families with neurogenic disorders. Most common mutation was miss-sense mutation^[51].

Rare genetic disorders affect millions of people across the globe either causing premature deaths or leaving them with prolonged co-morbidities. Genetic therapies are our first line revolutionary treatment options. Classic example is AAV (Aden-associated virus) gene therapy for Cystic Fibrosis^[52].

X-linked Retinitis Pigmentosa, which occurs due to mutation in RPGR gene, is another target for gene therapy. Retina is excellent for non-invasive procedures and it limits the immunological systemic spread^[53]. AAV vector mediated gene transfer in Hemophilia A and B can be used as a one-time treatment with factors level lasting for years^[54].

Use of more advance lentiviral vectors as gene therapy for Primary immunodeficiency, SCID, Wiskot-Aldrich and other Leukodystrophies has improved the biosafety. There have been promising results for the use of Autologous T-cells as an alternate strategy for Primary immunodeficiencies^[55]. Stem cell gene therapy for Fanconi's Anemia is another genetic approach that uses corrected stem cells to rapidly improve entire hematopoiesis of patient^[56]. SMN2 gene splicing modifiers like Nusinersen and Risdiplam, SMN1 gene replacement therapy with Zongelsma is used for Spinal muscular atrophy^[46].

CRISPR/CAS therapy is the gene therapy of future. It is going to be the therapy of choice for rare genetic diseases in the next 15–20 years. The technology is being tested for Thalassemia and Sickle cell Anemia, and is showing great potential^[57]. Thus, the use of genomic medicine, specifically the gene therapy is going to revolutionize the way we clinically diagnose and manage rare genetic disorders.

Pharmacogenomics

Pharmacogenomics leverages genomic biomarkers to predict individual responses to drug efficacy and toxicity. While factors like disease severity, diet, and other medications also influence drug responses, genetic differences significantly impact drug metabolism and action. Despite the growing body of research, replicating findings remains a challenge. Genome-wide association studies (GWAS) have identified genetic variations associated with psychiatric disorders and drug responses, but most findings lack consistent replication. The FDA includes pharmacogenomics information in drug labels, highlighting its growing recognition. The Clinical Pharmacogenetics Implementation Consortium (CPIC) aims to translate genetic data into clinical practice, providing guidelines for genome-informed prescribing of antidepressants and antipsychotics^[58].

Genetic factors significantly influence the metabolism of lamotrigine (LTG), an antiepileptic drug metabolized mainly by UGT enzymes, particularly UGT1A4. Polymorphisms in these enzymes, such as UGT1A4 and UGT2B7, can affect the drug's plasma concentration and efficacy. Additionally, genetic variations in transporters like OCT1 and ABCG2 also play a role in LTG pharmacokinetics, potentially necessitating dosage adjustments for effective treatment. Further research is needed to fully understand these genetic impacts and to optimize individual treatment plans^[59].

A 55-year-old clinical molecular geneticist became a patient after a tumor was detected, leading to a diagnosis of estrogen receptor positive breast cancer. Initially prescribed tamoxifen, she requested CYP2D6 testing due to concerns about genetic factors affecting the drug's efficacy. The test indicated an intermediate metabolizer status, prompting a switch to anastrozole, in line with CPIC guidelines^[60].

P2Y12 inhibitors like clopidogrel and prasugrel are metabolized into active forms by CYP enzymes, notably CYP2C19, which affects their efficacy. Carriers of CYP2C19*2 or 3 alleles, which reduce enzyme function, show decreased drug effectiveness and higher cardiovascular risks. Conversely, the CYP2C19*17 allele increases enzyme activity, enhancing drug efficacy and sometimes bleeding

risk. Other genetic variants, such as ABCB1 c.3435C>T and CES1, also influence drug metabolism but are not routinely tested^[61].

Genomic medicine and personalized treatment

This represents substantial improvements in customized care by offering tailored methods according to the specific genetic characteristics of everyone. One of the main advantages is increased diagnostic accuracy. Comprehending genetic differences allows clinicians to properly diagnose disorders that might otherwise be missed using conventional approaches, resulting in more accurate and earlier disease identification^[62]. Enhanced therapeutic efficacy and safety are also significant advantages. Personalized plans based on genomic data can assist in selecting the best medications and dosages, lowering adverse drug reactions and enhancing therapeutic success rates^[10]. This personalized approach ensures that therapies are both effective and safe for each patient. Furthermore, enhanced patient results and wellbeing are significant advantages. Individualized. Personalized treatments frequently result in better illness management and prognosis, which can benefit overall patient health and longevity^[12]. Patients benefit from therapies that are carefully designed to their unique genetics, resulting in speedier recovery times and an and a higher standard of living. However, putting genomic medicine to use offers its own set of obstacles. These include the requirement for vast genetic data, the complexities of interpreting genetic data and moral questions on genetic privacy and prejudice. Despite these difficulties that genomic medicine seems to be a promising topic regarding the future of individualized therapy.

Challenges in genomic medicine implementation

Technical and logistical challenges

Incorporating genomics into therapeutic practice requires a dependable bioinformatics infrastructure to manage and interpret vast datasets. This involves developing standardized procedures for the purpose of genome sequencing, analysis, and ensuring compatibility with existing electronic health records^[63]. Additionally, there is an urgent requirement for medical personnel to receive comprehensive training in genomics to effectively utilize these perspectives on patient care, bridging the gap between advanced technology and practical application^[64].

High costs and access issues

High price of genome sequencing and related technology is a major obstacle to its broad use, even though sequencing costs have dropped over time, it remains prohibitive for many healthcare systems and patients, particularly in low- and middle-income countries^[65].

However, ensuring equitable access to these treatments necessitates significant monetary commitment and supportive policies to subsidize costs and integrate genomic medicine into public healthcare systems.

Data privacy and ethical considerations

The gathering, storing, and use of genetic information raise substantial privacy concerns. Making sure patient data are securely stored and used in an ethical manner critical to maintaining public trust in genomic medicine^[66]. The ethical implications encompass preventing genetic discrimination and managing the potential psychological impact on patients who discover their risks. Rules and regulations must be established to protect people's genetic information and discuss the ethical ramifications of using genomic data^[67].

Privacy and data security

Handling and storage of genomic data

Regulating and storing genomic data presents tremendous privacy and security concerns since people's genetic information is considered sensitive. There is adequate legal protection for genomic data for clinical use, especially where the Health Information Portability and Accountability Act (HIPAA) applies. HIPAA outlines the degree of protection provided to such data and restricts access to only personnel in the clinical field. Some states have additional protections, but these vary from state to state, leading to disparities in privacy levels^[68].

Genomic data require robust protection from breaches, necessitating strong methods in their storage and transfer. Access to these data must be highly controlled, with monitoring of everyone who seeks access and logging of all actions on the data to properly identify violators. Preventive measures for the protection of genomic data are of utmost importance due to the severe consequences individuals may face from the misuse of their information^[68].

Consent and confidentiality issues

Two critical aspects of handling genomic data are consent and confidentiality, aligning with patients' concerns about the privacy of their genetic sequences and potential misuse. The Genetic Information Nondiscrimination Act (GINA), signed into law in 2008, addresses discrimination in insurance and employment based on genetic characteristics but does not cover life, disability, or long-term care insurance.

Privacy is paramount; each patient must know how their genetic details will be utilized, where they will be stored, and who will be allowed access. The HIPAA Privacy Rule generally restricts the disclosure of genetic data without the patient's consent, though there may be exceptional cases requiring the disclosure to at-risk relatives based on ethical principles. For instance, physicians might encourage patients to disclose genetic risks to their families while respecting patient privacy and legal guidelines^[68].

Disparities in access to genomic testing and personalized treatments

Lack of equal access to genomic testing and personalized treatments is rampant, with minorities, women, rural patients, uninsured/underinsured patients and those with low education and income levels being most affected. For instance, in the case of

breast cancer, non-Hispanic Black women receive low rates of BRCA testing compared to non-Hispanic White women. This is partly because there are fewer conversations about genetic testing with healthcare providers and fewer referrals to genetic counselors among minority-serving physicians. Therefore, these disparities also translate to preventive measures, such as lower risk-reducing surgeries among Black women and fewer cases of cascade screening among Black families with BRCA variants. Furthermore, the underserved population, including racial and ethnic minorities, low-income groups, and women, have barriers in accessing treatments such as PCSK9 inhibitors for FH leading to poor cholesterol control and poor health outcomes^[69].

Strategies to improve access and reduce inequities

In order to improve health equity for genomic medicine it is necessary to engage participants from non-European decent and other deprived population backgrounds. Increasing rates of utilization of genomic services depends on the ability to make such tests accessible and have acceptable coverage in various settings such as community hospitals or primary care physicians' offices. Training of the workforce and infrastructure improvement in MSIs aids in improving culturally sensitive care and research. Sparking collaboration with the local communities and the healthcare providers ensures a mutual understanding between the two that will make genomic research to reflect their perception. Furthermore, the financing of the research facilities in other than academic institutions and in underprivileged regions contributes to a wider deployment's and participants' integration. Together, these strategies seek to address health disparity and guarantee the equitable improvement of all people through genomic medicine^[70].

Possible threats and legal remedies

Genetic discrimination focuses on prejudice against people with specific genetic characteristics which exposes them to serious threats such as loss of insurance, inability to secure a job and social exclusion. The following risks have however been regulated by law especially by the Genetic Information Non-discrimination Act of 2008. GINA offers certain federal anti-discrimination provisions for genetic tests whether from the state's and health insurers' or employers' perspectives as they ban such entities from obtaining or using genetic information for underwriting purposes or employment respectively^[71]. However, GINA does not cover life, disability or long-term care insurance and for this void state laws try to provide a solution^[71].

Measures to prevent discrimination

Some recommendations to address discrimination in genomic medicine are having general statutes such as the Genetic Information Non-discrimination Act, enacted with an aim of preventing discrimination of individuals based on genetic information. In employment and health insurance decisions, GINA is enforced; however, health insurers cannot use genetic information to underwrite life, disability, or long-term care insurance. Furthermore, there are intentions to increase the consciousness about the protection against genetic non-

discrimination and to remove the general distrust thereby impeding genomic research due to apprehension for discrimination. Furthermore, the Affordable Care Act (ACA) has also sought to fill gaps by extending provisions that ban the health insurance status discrimination based on pre-existing conditions to include genetic information; thereby supporting GINA's provisions aimed at addressing employment discrimination. Enlarging such protections to encompass all sorts of insurance and ensuring people advocate for a similar system that combines risk might help to avoid discrimination and promote the proper usage of genomic medicine further^[72].

Future directions and emerging trends

Advancements in genomic technologies

New sequencing techniques and technologies. Since the Human Genome Project's completion in 2003, DNA sequencing technologies have advanced significantly to fill previously existing gaps^[73]. There are primarily two types of DNA sequencing technologies: short-read sequencing and long-read sequencing. Short-read sequencing methods, such as sequence molecule fluorescent sequencing and single-molecule nanopore base sequencing, generate genetic information in 100–300 base pairs per read^[19]. They are efficient and cost-effective but often miss repetitive regions, duplicated sequences, and complex structural variants, leaving gaps in the data. While long-read sequencing provides better resolution of complex regions and structural variants, it is typically more expensive and has higher error rates. Combining both approaches can enhance genomic analysis^[73].

Innovations in bioinformatics and data integration. Advancements in pharmacogenomics and the integration of sequenced genomes with medical records, expression profiles, and imaging studies necessitate robust data storage solutions like cloud computing. It is crucial to manage these data while ensuring both accessibility and confidentiality. In the realm of AI, such comprehensive data can significantly enhance the development of genomics and improve outcomes. However, it is vital to apply this knowledge and data judiciously in clinical settings to ensure its effectiveness and ethical use^[74,75].

Expanding the scope of personalized medicine

Integration of multi-omics data. Multi-omics refers to the use of multiple biological "omes" such as genome, proteome, transcriptome, epigenome, metabolome, radiomics, and microbiome to provide data to achieve a holistic understanding of biological systems and enhance personalized medical treatments^[76]. Multiomics can provide the missing link of information in the study of genomics and help uncover the pathophysiology underlying a disease which will help provide a new approach to its detection, treatment, and prevention^[77]. This new approach will pave the way for personalized medicine and optimize its clinical outcome based on the uniqueness of an individual^[9].

Multi-omics approaches can fill critical gaps in genomic research by providing comprehensive insights into the underlying mechanisms of diseases. By integrating various types of omics data, such as genomics, proteomics, and metabolomics, researchers can better understand disease pathophysiology. This enhanced understanding enables the development of novel strategies for disease detection, treatment, and prevention^[77]. The application of these strategies will support the advancement of personalized medicine, which aims to tailor medical interventions to the unique characteristics of each individual. Ultimately, this personalized approach will optimize clinical outcomes and improve patient care by addressing the specific needs and conditions of each patient, leading to more effective and targeted treatments^[9].

Broader applications across different diseases and population. Precision medicine can categorize individuals based on their clinical features, treatment responses, and prognostic factors^[78]. By leveraging multi-omics studies in diseases such as inflammatory bowel disease, various cancers, and lifestyle-related conditions like diabetes, personalized medicine aims to tailor treatments to each person's unique profile. Since pharmacokinetics is closely linked to genetic variations, personalized medicine has the potential to revolutionize genomics and drive the development of new therapies. This approach integrates comprehensive omics data to refine treatment strategies, enhancing their effectiveness and leading to more targeted, individualized healthcare solutions. Ultimately, this method supports more precise and effective management of diverse health conditions, contributing to advancements in medical science and patient care^[9].

Role of AI in precision medicine. Artificial intelligence excels at processing multidimensional clinical and biological data, which is critical for precision medicine. It assists in discovering biomarkers via genetic sequencing and other data sources, turning complex data into meaningful insights for tailored treatment strategies^[79]. Artificial intelligence algorithms, such as machine learning and deep learning, improve disease diagnosis and early detection. This is especially visible in disciplines like oncology and cardiovascular care, where AI helps anticipate disease risk and stratify individuals based on their unique traits^[80]. AI gives clinicians additional insight by combining data from many sources, such as electronic health records (EHR), imaging data, and omics data. This integration aids clinical decision-making by increasing the accuracy of diagnoses and the efficacy of treatment strategies^[81]. AI is widely employed in oncology for tasks such as tumor identification, therapy planning, and prognosis prediction. It assists in identifying new biomarkers and understanding tumor heterogeneity, resulting in more accurate and effective cancer treatments^[82]. AI helps in diagnosis and forecast the prognosis of cardiovascular illnesses. It employs several machine learning models to assess data from EHR, imaging and omics thereby boosting the accuracy of risk prediction and treatment planning^[83]. AI has the potential to predict the risks and outcomes of neurodevelopmental diseases by examining genomic variants and other biological markers. However the complexity and variability of

these illnesses provide substantial hurdles that AI continues to solve^[84].

Clinical trials on genomics and personalized treatment. Personalized medicine in oncology is adapting treatment to individual patient features, especially genomic and molecular markers. This strategy seeks to give the "right treatment for the right person at the right time" by using genetic information to guide therapeutic decisions^[85].

The MINDACT study investigated the use of a 70-gene signature to inform chemotherapy decisions in early-stage breast cancer. A decision-analytic modeling technique indicates that fewer women may benefit from genomic testing and treatment than previously indicated by the trial, underlining the necessity for personalized decision-making based on genomic risk^[86].

The PROMISE study finds that concentrations of hs-cTn and IL-6 were associated with coronary artery disease (CAD) characteristics and major adverse cardiovascular events (MACEs), indicating that myocardial injury and inflammation play a role in CAD pathophysiology. This association was strongest in participants with non-obstructive CAD, highlighting an opportunity to tailor treatment for this at-risk group^[87].

Intensive blood pressure management in older hypertensive persons with sarcopenia was related with a lower risk of cardiovascular disease (CVD) without an increased risk for adverse events, suggesting potential for individualized treatment techniques targeted to this at-risk group^[88].

STK11/LKB1 mutations were discovered to be a prominent cause of primary resistance to PD-1 inhibitors in KRAS-mutant lung adenocarcinoma (LUAC). This resistance was demonstrated in numerous clinical cohorts, with different response to PD-1 blocking among LUAC subtypes. These findings suggest that STK11/LKB1 mutations can be employed as a predictive biomarker for PD-1 inhibitor efficacy potentially informing customized treatment options for KRAS-mutant patients^[89].

KRAS codon G12 mutations have been identified as biomarkers of resistance to trifludine/tipiracil (FTD/TPI) chemotherapy in metastatic colorectal cancer (mCRC), with patients carrying these mutations showing significantly reduced overall survival benefit from treatment, implying that genomics-based precision medicine could inform chemotherapy selection and improve outcomes for mCRC^[90].

The potential of genomics-based tailored treatment, demonstrating that magnesium supplementation can regulate DNA methylation in the TMPRSS2 gene, which is critical for SARS-CoV-2 viral entry. Adjusting magnesium levels in individuals with specific calcium-to-magnesium intake ratios suggests a novel gene-environment interaction that could be leveraged for personalized prevention strategies and treatment of early COVID-19, potentially altering viral susceptibility based on individual genetic and nutritional factors.^[91]

Conclusion

Genomic medicine has increased our knowledge of genetic variations, resulting in more accurate diagnosis and personalized

Khan et al. Annals of Medicine & Surgery (2025) treatment^[18]. DNA sequencing technology, genetic data integration into clinical care, and the use of multi-omics techniques are among the most significant developments.

Future research should focus on increasing access to genetic technology, tackling ethical challenges, and enhancing bioinformatics facilities. Clinical practice needs to change to include these developments, providing fair and efficient individual treatment.

As genomic medicine growing, it will play an increasingly significant part in transforming healthcare. Addressing existing challenges will be important for achieving its full assurance, leading to more customized, precise, and efficient treatments that improves outcomes for patients.

Ethical approval

Ethical approval was not required for this review.

Consent

Informed consent was not required for this review.

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Author's contribution

A.K.: conception and design of the study, drafting the manuscript, critical revision of the article for important intellectual content, and final approval of the version to be published. A.R. B.: acquisition of data, analysis and interpretation of data, drafting sections of the manuscript, and revising it critically for important intellectual content. N.B. and C.S.P.: acquisition of data, drafting sections of the manuscript, revising it critically for important intellectual content, and providing final approval of the version to be published. J.D.: assistance in data collection, drafting sections of the manuscript, and revising it critically for important intellectual content. M.G. and S.P.: data interpretation, drafting sections of the manuscript, and revising it critically for important intellectual content. K.G. and M.B.: analysis and interpretation of data, drafting sections of the manuscript, and revising it critically for important intellectual content. S.N.: data collection and interpretation, drafting sections of the manuscript, and revising it critically for important intellectual content. U.H.: assistance in data collection, drafting sections of the manuscript, and revising it critically for important intellectual content. S.P.K. and A.R.P.: analysis and interpretation of data, drafting sections of the manuscript, and revising it critically for important intellectual content. U.J.: data collection, drafting sections of the manuscript, and revising it critically for important intellectual content.

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References

- [1] Nielsen FC. Genomic medicine. *Ugeskr Laeger* 2019;181:V01190031.
- [2] Musunuru K, Hickey KT, Al-Khatib SM, et al. Basic concepts and potential applications of genetics and genomics for cardiovascular and stroke clinicians. *Circulation* 2015;8:216–42.
- [3] Watson J, Crick F. Molecular structure of nucleic acids: a structure for deoxyribose nucleic acid. *Nature* 1953;171:737–38.
- [4] Sanger F, Thompson EOP. The amino-acid sequence in the glycol chain of insulin. 2. The investigation of peptides from enzymic hydrolysates. *Biochem J* 1953;53:366–74.
- [5] Maria Giani A, Roberto Gallo G, Gianfranceschi L, et al. Long walk to genomics: history and current approaches to genome sequencing and assembly. *Comput Struct Biotechnol J* 2020;18:9–19.
- [6] Staden R. A strategy of DNA sequencing employing computer programs. *Nucleic Acids Res* 1979;6:2601–10.
- [7] Baer R, Bankier A, Biggin M, et al. DNA sequence and expression of the B95-8 Epstein-Barr virus genome. *Nature* 1984;310:207–11.
- [8] Heather JM, Chain B. The sequence of sequencers: the history of sequencing DNA. *Genomics* 2016;107:1–8.
- [9] Goetz LH, Schork NJ. Personalized medicine: motivation, challenges, and progress. *Fertil Steril* 2018;109:952–63.
- [10] Alzu'bi AA, Zhou L, Watzlaf VJM. Genetic variations and precision medicine. *Perspect Health Inf Manag* 2019;16:1a.
- [11] Bustin M, Misteli T. Nongenetic functions of the genome. *Science* 2016;352:aad6933.
- [12] Juengst E, McGowan M. Why does the shift from 'personalized medicine' to 'precision health' and 'wellness genomics' matter? *AMA J Ethics* 2018;20:e881–e890.
- [13] Shastri BS. SNPs: impact on gene function and phenotype. *Methods Mol Biol* 2009;578:3–22.
- [14] Say YH. The association of insertions/deletions (INDELs) and variable number tandem repeats (VNTRs) with obesity and its related traits and complications. *J Physiol Anthropol* 2017;36:25.
- [15] Shaikh TH, Gai X, Perin JC, et al. High-resolution mapping and analysis of copy number variations in the human genome: a data resource for clinical and research applications. *Genome Res* 2009;19:1682–90.
- [16] Lerman C, Schnoll RA, Lw H Jr, et al. Use of the nicotine metabolite ratio as a genetically informed biomarker of response to nicotine patch or varenicline for smoking cessation: a randomised, double-blind placebo-controlled trial. *Lancet Respir Med* 2015;3:131–38.
- [17] Manolio TA, Chisholm RL, Ozenberger B, et al. Implementing genomic medicine in the clinic: the future is here. *Genet Med* 2013;15:258–67.
- [18] Manolio TA, Rowley R, Williams MS, et al. Opportunities, resources, and techniques for implementing genomics in clinical care. *Lancet* 2019;394:511–20.
- [19] McCombie WR, McPherson JD, Mardis ER. Next-generation sequencing technologies. *Cold Spring Harb Perspect Med* 2019;9:a036798.
- [20] Ishida C, Zubair M, Gupta V. Molecular Genetics Testing. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2024.
- [21] Chau MHK, Choy KW. The role of chromosomal microarray and exome sequencing in prenatal diagnosis. *Curr Opin Obstet Gynecol* 2021;33:148–55.
- [22] Zhang S, Xu Y, Lu D, et al. Combined use of karyotyping and copy number variation sequencing technology in prenatal diagnosis. *PeerJ* 2022;10:e14400.
- [23] Chrzanowska NM, Kowalewski J, Lewandowska MA. Use of Fluorescence In Situ Hybridization (FISH) in diagnosis and tailored therapies in solid tumors. *Molecules* 2020;25:1864.

- [24] Russo J, Giri VN. Germline testing and genetic counselling in prostate cancer. *Nat Rev Urol* 2022;19:331–43.
- [25] Duncavage EJ, Bagg A, Hassarjian RP, *et al.* Genomic profiling for clinical decision making in myeloid neoplasms and acute leukemia. *Blood* 2022;140:2228–47.
- [26] Striano P, Minassian BA. From genetic testing to precision medicine in epilepsy. *Neurotherapeutics* 2020;17:609–15.
- [27] Zou YB, Hui RT, Song L. The era of clinical application of gene diagnosis in cardiovascular diseases is coming. *Chronic Dis Transl Med* 2019;5:214–20.
- [28] Westerman K, Sebastiani P, Jacques P, *et al.* DNA methylation modules associate with incident cardiovascular disease and cumulative risk factor exposure. *Clin Epigenetics* 2019;11:142.
- [29] Zhu H, Wang X, Meng X, *et al.* Selenium supplementation improved cardiac functions by suppressing DNMT2-mediated GPX1 promoter DNA methylation in AGE-induced heart failure. *Oxid Med Cell Longev* 2022;2022:5402997.
- [30] Kim KH, Pereira NL. Genetics of cardiomyopathy: clinical and mechanistic implications for heart failure. *Korean Circ J* 2021;51:797–836.
- [31] Ontario Health (Quality). Genetic testing for familial hypercholesterolemia: health technology assessment. *Ont Health Technol Assess Ser* 2022;22:1–155.
- [32] Brown EE, Sturm AC, Cuchel M, *et al.* Genetic testing in dyslipidemia: a scientific statement from the National Lipid Association. *J Clin Lipidol* 2020;14:398–413.
- [33] Rincon MY, VandenDriessche T, Chuah MK. Gene therapy for cardiovascular disease: advances in vector development, targeting, and delivery for clinical translation. *Cardiovasc Res* 2015;108:4–20.
- [34] Ferrara K, Pollard R, Borden M. Ultrasound microbubble contrast agents: fundamentals and application to gene and drug delivery. *Annu Rev Biomed Eng* 2007;9:415–47.
- [35] Gill SL, O'Neill H, McCoy RJ, *et al.* Enhanced delivery of microRNA mimics to cardiomyocytes using ultrasound responsive microbubbles reverses hypertrophy in an in-vitro model. *Technol Healthc* 2014;22:37–51.
- [36] Naito Y, Aburatani H, Amano T, *et al.* Clinical practice guidance for next-generation sequencing in cancer diagnosis and treatment (edition 2.1). *Int J Clin Oncol* 2021;26:233–83.
- [37] Swain SM, Shastry M, Hamilton E. Targeting HER2-positive breast cancer: advances and future directions. *Nat Rev Drug Discovery* 2023;22:101–26.
- [38] Fu K, Xie F, Wang F, *et al.* Therapeutic strategies for EGFR-mutated non-small cell lung cancer patients with osimertinib resistance. *J Hematol Oncol* 2022;15:173.
- [39] Hughes T, Klairmont M, Sharfman WH, *et al.* Interleukin-2, Ipilimumab, and Anti-PD-1: clinical management and the evolving role of immunotherapy for the treatment of patients with metastatic melanoma. *Cancer Biol Ther* 2021;22:513–26.
- [40] Yayan J. Denosumab for effective tumor size reduction in patients with giant cell tumors of the bone: a systematic review and meta-analysis. *Cancer Control* 2020;27:107327482093482.
- [41] Maggiore U, Palmisano A, Buti S, *et al.* Chemotherapy, targeted therapy and immunotherapy: which drugs can be safely used in the solid organ transplant recipients? *Transplant Int* 2021;34:2442–58.
- [42] Onozawa H, Saito H, Sunami K, *et al.* Lung adenocarcinoma in a patient with a cis-based profiling test: negative diagnosis on initial companion test and successful treatment with osimertinib. *Thorac Cancer* 2020;11:3599–604.
- [43] Aigner B, Plötz SG, Schaller G. Triple-negative breast cancer possibly transforming into malignant melanoma due to targeted therapy? A case report and review of literature. *Wien Med Wochenschr* 2013;163:495–98.
- [44] Hamilton MP, Sugio T, Noordenbos T, *et al.* Risk of second tumors and T-cell lymphoma after CAR T-cell therapy. *N Engl J Med* 2024;390: 2047–60.
- [45] Jiang Y, Zhang H, Wang J, *et al.* Targeting extracellular matrix stiffness and mechanotransducers to improve cancer therapy. *J Hematol Oncol* 2022;15:34.
- [46] Ontario Health (Quality). Carrier screening programs for cystic fibrosis, fragile X syndrome, hemoglobinopathies and thalassemia, and spinal muscular atrophy: a health technology assessment. *Ont Health Technol Assess Ser* 2023;23:1–398.
- [47] Szustakowski JD, Balasubramanian S, Kvikstad E, *et al.* Advancing human genetics research and drug discovery through exome sequencing of the UK Biobank. *Nat Genet* 2021;53:942–48.
- [48] Bycroft C, Freeman C, Petkova D, *et al.* The UK Biobank resource with deep phenotyping and genomic data. *Nature* 2018;562:203–09.
- [49] Johns DR. Seminars in medicine of the Beth Israel Hospital, Boston. Mitochondrial DNA and disease. *N Engl J Med* 1995;333:638–44.
- [50] Nair P, Sabbagh S, Mansour H, *et al.* Contribution of next generation sequencing in pediatric practice in Lebanon. A study on 213 cases. *Mol Genet Genomic Med* 2018;6:1041–52.
- [51] Alazami AM, Patel N, Shamseldin HE, *et al.* Accelerating novel candidate gene discovery in neurogenetic disorders via whole-exome sequencing of prescreened multiplex consanguineous families. *Cell Rep* 2015;10:148–61.
- [52] Guggino WB, Cebotaru L. Adeno-associated virus (AAV) gene therapy for cystic fibrosis: current barriers and recent developments. *Expert Opin Biol Ther* 2017;17:1265–73.
- [53] Martinez-Fernandez De La Camara C, Nanda A, Salvetti AP, *et al.* Gene therapy for the treatment of X-linked retinitis pigmentosa. *Expert Opin Orphan Drugs* 2018;6:167–77.
- [54] Reiss UM, Zhang L, Ohmori T. Hemophilia gene therapy-new country initiatives. *Haemophilia* 2021;27:132–41.
- [55] Booth C, Romano R, Roncarolo MG, *et al.* Gene therapy for primary immunodeficiency. *Hum Mol Genet* 2019;28:R15–R23.
- [56] Hanenberg H, Roellecke K, Wiek C. Stem cell genetic therapy for fanconi anemia - a new hope. *Curr Gene Ther* 2017;16:309–20.
- [57] Braga LAM, Conte Filho CG, Mota FB. Future of genetic therapies for rare genetic diseases: what to expect for the next 15 years? *Ther Adv Rare Dis* 2022;3:26330040221100840.
- [58] Koromina M, Koutsilieri S, Patrinos GP. Delineating significant genome-wide associations of variants with antipsychotic and antidepressant treatment response: implications for clinical pharmacogenomics. *Hum Genomics* 2020;14:4.
- [59] Zhao W, Meng H. Effects of genetic polymorphism of drug-metabolizing enzymes on the plasma concentrations of antiepileptic drugs in Chinese population. *Bioengineered* 2022;13:7709–45.
- [60] Luzum JA, Petry N, Taylor AK, *et al.* Moving pharmacogenetics into practice: it's all about the evidence! *Clin Pharmacol Ther* 2021;110: 649–61.
- [61] Božina N, Vrkić Kirhmajer M, Šimičević L, *et al.* Use of pharmacogenomics in elderly patients treated for cardiovascular diseases. *Croat Med J* 2020;61:147–58.
- [62] Weissenbach J. The rise of genomics. *C R Biol* 2016;339:231–39.
- [63] Manolio TA, Chisholm RL, Ozenberger B, *et al.* Implementing genomic medicine in the clinic: the future is here. *Genet Med* 2017;19:733–40.
- [64] Evans JP, Powell BC, Berg JS. Finding the rare pathogenic variants in a human genome. *JAMA* 2017;317:1904–05.
- [65] Adams DR, Eng CM. Next-generation sequencing to diagnose suspected genetic disorders. *N Engl J Med* 2018;379:1353–62.
- [66] Terry SF. Precision medicine: ethics, politics, and the promise of a good life. *Hastings Cent Rep* 2015;45:14–17.
- [67] Klitzman R. Views and experiences of IRBs concerning research integrity. *J Law Med Ethics* 2010;38:194–207.
- [68] Stiles D, Appelbaum PS. Cases in precision medicine: concerns about privacy and discrimination after genomic sequencing. *Ann Intern Med* 2019;170:717–21.
- [69] Khoury MJ, Bowen S, Dotson WD, *et al.* Health equity in the implementation of genomics and precision medicine: a public health imperative. *Genet Med* 2022;24:1630–39.
- [70] Joama S, Hahn MJ, Hindorff LA, *et al.* Defining and achieving health equity in genomic medicine. *Ethn Dis* 2019;29:173–78.
- [71] Miller AR, Tucker C. Privacy protection, personalized medicine, and genetic testing. *Manag Sci* 2018;64:4648–68.
- [72] Green RC, Lautenbach D, McGuire AL. GINA, genetic discrimination, and genomic medicine. *N Engl J Med* 2015;372:397–99.
- [73] Warburton PE, Sebra RP. Long-read DNA sequencing: recent advances and remaining challenges. *Annu Rev Genomics Hum Genet* 2023;24:109–32.
- [74] Prokop JW, May T, Strong K, *et al.* Genome sequencing in the clinic: the past, present, and future of genomic medicine. *Physiol Genomics* 2018;50:563–79.
- [75] Liu Y, Chen Y, Han L. Bioinformatics: advancing biomedical discovery and innovation in the era of big data and artificial intelligence. *Innov Med* 2023;1:100012.

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- [76] Chen C, Wang J, Pan D, *et al.* Applications of multi-omics analysis in human diseases. *MedComm* (2020) 2023;4:e315.
- [77] Sun YV, Hu YJ. Integrative analysis of multi-omics data for discovery and functional studies of complex human diseases. *Adv Genet* 2016;93:147–90.
- [78] Jagirdhar GSK, Perez JA, Perez AB, *et al.* Integration and implementation of precision medicine in the multifaceted inflammatory bowel disease. *World J Gastroenterol* 2023;29:5211–25.
- [79] Mumtaz H, Saqib M, Jabeen S, *et al.* Exploring alternative approaches to precision medicine through genomics and artificial intelligence – a systematic review. *Front Med (Lausanne)* 2023;10:1227168.
- [80] Johnson K, Wei W, Weeraratne D, *et al.* Precision medicine, AI, and the future of personalized health care. *Clin Transl Sci* 2021;14:86–93.
- [81] Hamamoto R, Suvarna K, Yamada M, *et al.* Application of artificial intelligence technology in oncology: towards the establishment of precision medicine. *Cancers (Basel)* 2020;12:3532.
- [82] Mohsen F, Al-Saadi B, Abdi N, *et al.* Artificial intelligence-based methods for precision cardiovascular medicine. *J Pers Med* 2023;13:1268.
- [83] Uddin M, Wang Y, Woodbury-Smith M. Artificial intelligence for precision medicine in neurodevelopmental disorders. *NPJ Digital Med* 2019;2:112.
- [84] Iqbal J, Cortés Jaimes DC, Makineni P, *et al.* Reimagining healthcare: unleashing the power of artificial intelligence in medicine. *Cureus* 2023;15:e44658.
- [85] Superchi C, Bouvier F, Gerardi C, *et al.* Study designs for clinical trials applied to personalised medicine: a scoping review. *BMJ Open* 2022;12:e052926.
- [86] Steyerberg E, De Wreede L, Van Klaveren D, *et al.* Personalized decision making on genomic testing in early breast cancer: expanding the MINDACT trial with decision-analytic modeling. *Med Decis Making* 2021;41:354–65.
- [87] Ferencik M, Mayrhofer T, Lu MT, *et al.* Coronary atherosclerosis, cardiac troponin, and interleukin-6 in patients with chest pain: the PROMISE trial results. *JACC Cardiovasc Imaging* 2022;15:1427–38.
- [88] Mirzai S, Persits I, Kazibwe R, *et al.* Relationship between sarcopenia and intensive blood pressure control efficacy and safety: a secondary analysis of SPRINT. *Hypertension* 2024;81:e77–e87.
- [89] Skoulidis F, Goldberg ME, Greenawalt DM, *et al.* STK11/LKB1 mutations and PD-1 inhibitor resistance in KRAS-mutant lung adenocarcinoma. *Cancer Discov* 2018;8:822–35.
- [90] van de Haar J, Ma X, Ooft SN, *et al.* Codon-specific KRAS mutations predict survival benefit of trifluridine/tipiracil in metastatic colorectal cancer. *Nat Med* 2023;29:605–14.
- [91] Fan L, Zhu X, Zheng Y, *et al.* Magnesium treatment on methylation changes of transmembrane serine protease 2 (TMPRSS2). *Nutrition* 2021;89:111340.